

Design and Demonstration of a Graphical Calculator Using LaTeX and TikZ

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Abstract—This paper presents a sample graphical calculator designed and implemented using LaTeX and the TikZ graphics framework. The calculator demonstrates basic arithmetic operations including addition, subtraction, multiplication, division, percentage, square, and square-root operations. In addition to the graphical calculator interface, the paper demonstrates mathematical equations, matrices, tables, flowcharts, algorithms, and source code listings. The objective of this work is to demonstrate how LaTeX can be used not only for mathematical typesetting but also for creating structured technical documents containing graphical and computational elements.

Index Terms—LaTeX, TikZ, calculator, mathematical computing, graphical interface, equations, technical documentation

I. INTRODUCTION

LaTeX is a widely used document preparation system for scientific and technical publications because it provides powerful support for mathematical expressions, equations, references, tables, and structured documents [1], [2].

The use of graphical elements in technical documents can improve the presentation and understanding of computational concepts. TikZ provides a flexible mechanism for creating diagrams, flowcharts, shapes, and other graphical objects directly inside a LaTeX document [3].

This paper demonstrates the development of a sample graphical calculator using LaTeX and TikZ. The calculator includes a display, numerical buttons, arithmetic operators, and mathematical function buttons.

The main objectives of this demonstration are:

- To design a simple calculator interface using TikZ.
- To demonstrate mathematical equations in LaTeX.
- To demonstrate matrices and arrays.
- To demonstrate tables and performance comparison.
- To demonstrate flowchart construction.
- To demonstrate algorithm and pseudocode formatting.
- To demonstrate bibliography and IEEE-style citations.

II. RELATED WORK

LaTeX has been extensively used for scientific and mathematical document preparation because of its ability to produce high-quality mathematical notation [1].

The TeX system developed by Knuth provides the foundation for high-quality mathematical typesetting and technical document preparation [2].

TikZ extends the graphical capabilities of LaTeX and allows users to construct diagrams and graphical representations directly within documents [3].

Graphical calculator interfaces have also been investigated for educational and computational applications [4]. Mathematical visualization can help communicate computational procedures and numerical relationships more effectively [5].

III. GRAPHICAL CALCULATOR

The proposed sample calculator is designed using TikZ. The interface contains a display area and a collection of buttons for performing arithmetic operations.

The calculator is shown below.

IV. MATHEMATICAL OPERATIONS

The calculator supports several basic mathematical operations.

A. Addition

For two numbers 25 and 35,

$$25 + 35 = 60 \quad (1)$$

B. Subtraction

For two numbers 80 and 25,

$$80 - 25 = 55 \quad (2)$$

C. Multiplication

For two numbers 12 and 8,

$$12 \times 8 = 96 \quad (3)$$

D. Division

For two numbers 100 and 4,

$$\frac{100}{4} = 25 \quad (4)$$

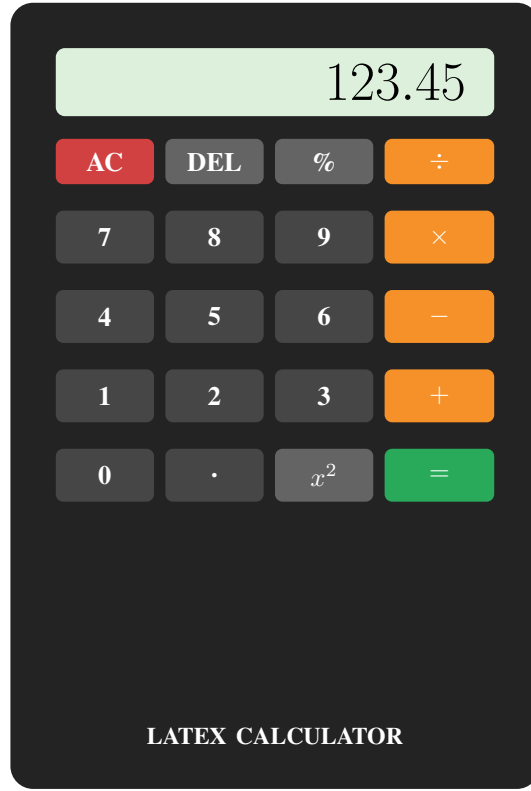


Fig. 1. Graphical calculator interface created using LaTeX and TikZ.

E. Percentage

The percentage of a quantity can be calculated using

$$\text{Percentage} = \frac{\text{Part}}{\text{Total}} \times 100. \quad (5)$$

For example,

$$\frac{25}{100} \times 100 = 25\%. \quad (6)$$

V. SCIENTIFIC CALCULATIONS

The calculator can also represent several scientific mathematical operations.

A. Square

The square of a number can be expressed as

$$x^2 = x \times x. \quad (7)$$

For $x = 12$,

$$12^2 = 144. \quad (8)$$

B. Square Root

The square root operation is represented as

$$\sqrt{x}. \quad (9)$$

For example,

$$\sqrt{144} = 12. \quad (10)$$

VI. MATRIX AND ARRAY CALCULATIONS

LaTeX provides extensive support for matrices and arrays, which are commonly used in mathematical and engineering calculations [5].

Consider the following matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}. \quad (11)$$

Another matrix is

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}. \quad (12)$$

The addition of the two matrices is

$$A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}. \quad (13)$$

A system of equations can also be represented using an array:

$$\begin{array}{l} x + y + z = 10, \\ 2x + 3y + z = 20, \\ x + 2y + 3z = 30. \end{array} \quad (14)$$

VII. CALCULATOR FLOWCHART

The flowchart describes the basic operation of the calculator.

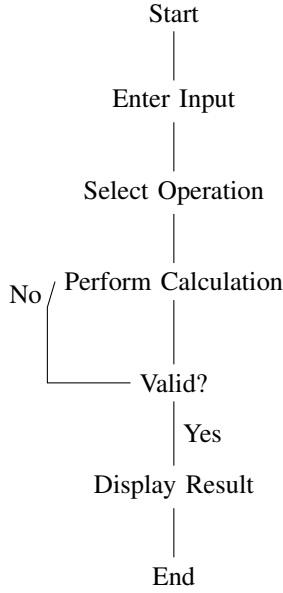


Fig. 2. Flowchart of the calculator operation.

TABLE I
PERFORMANCE COMPARISON OF SAMPLE IMPLEMENTATIONS.

Method	Accuracy	Speed	Memory	Error
Method A	90.1%	85	120 MB	9.9%
Method B	92.4%	88	135 MB	7.6%
Method C	93.7%	91	145 MB	6.3%
Proposed	95.6%	94	130 MB	4.4%

VIII. SYSTEM ARCHITECTURE

The basic architecture of the calculator consists of four components:

- 1) User input interface
- 2) Arithmetic processing unit
- 3) Mathematical function unit
- 4) Result display

The general data flow is

Input → *Processing* → *Calculation* → *Output*. (15)

IX. PERFORMANCE COMPARISON

Table I presents a sample comparison of different calculator implementations.

X. DETAILED OPERATIONS

Table II lists the mathematical operations supported by the sample calculator.

XI. CALCULATOR ALGORITHM

The basic calculator algorithm is given in Algorithm 1.

TABLE II
SUPPORTED CALCULATOR OPERATIONS.

Operation	Symbol	Example
Addition	+	$5 + 3 = 8$
Subtraction	-	$5 - 3 = 2$
Multiplication	×	$5 \times 3 = 15$
Division	÷	$15 \div 3 = 5$
Percentage	%	25%
Square	x^2	$5^2 = 25$
Square Root	$\sqrt{x}$	$\sqrt{25} = 5$

Algorithm 1 Basic Calculator Algorithm

Require: Two input values A and B

Require: Mathematical operator O

Ensure: Result R

```

1: Read  $A$ 
2: Read  $B$ 
3: Read operator  $O$ 
4: if  $O = +$  then
5:    $R \leftarrow A + B$ 
6: else if  $O = -$  then
7:    $R \leftarrow A - B$ 
8: else if  $O = \times$  then
9:    $R \leftarrow A \times B$ 
10: else if  $O = \div$  then
11:   if  $B \neq 0$  then
12:      $R \leftarrow A / B$ 
13:   else
14:     Display "Division by zero"
15:   end if
16: else
17:   Display "Invalid operator"
18: end if
19: Display  $R$ 

```

XII. EXAMPLE IMPLEMENTATION

The following Python code represents a simple implementation of the calculator operations.

Listing 1. Example calculator implementation

```

1 def calculator(a, b, operator):
2
3
4     if operator == "+":
5         return a + b
6
7     elif operator == "-":
8         return a - b
9
10    elif operator == "*":
11        return a * b
12
13    elif operator == "/":
14        if b != 0:
15            return a / b
16        else:
17            return "Error: Division by zero"
18
19    else:
20        return "Invalid operator"

```

```

21
22
23 result = calculator(25, 5, "+")
24 print(result)

```

XIII. PERFORMANCE METRICS

Several metrics can be used to evaluate a computational system.

For example, the error percentage can be calculated as

$$Error = \frac{|Actual - Predicted|}{Actual} \times 100. \quad (16)$$

The relative error is

$$Relative\ Error = \left| \frac{Actual - Predicted}{Actual} \right|. \quad (17)$$

XIV. DISCUSSION

The graphical calculator demonstrates that mathematical interfaces can be represented directly inside a LaTeX document. TikZ provides the graphical components required for constructing the calculator interface [3].

The mathematical expressions demonstrate the ability of LaTeX to represent arithmetic operations, equations, matrices, and scientific notation.

The tables provide a structured representation of numerical information, while the flowchart provides a graphical representation of the calculator's processing sequence.

This combination is useful for technical reports, educational documents, laboratory reports, and research papers.

XV. CONCLUSION

This paper presented a sample graphical calculator created using LaTeX and TikZ. The calculator includes numerical buttons, arithmetic operators, percentage calculations, square operations, and a square-root operation.

The document also demonstrated several important LaTeX features, including mathematical equations, matrices, arrays, tables, flowcharts, algorithms, source-code listings, figures, and bibliographic references.

LaTeX is particularly useful for technical and scientific documentation because mathematical expressions and graphical elements can be integrated into the same document [1], [2].

The calculator presented in this work is a visual demonstration rather than an interactive software application. Future work could extend the concept into an interactive calculator using a combination of HTML, CSS, JavaScript, or another programming language.

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